...questions 1,2 consider the current through a 1 mH inductor is $i(t)=60\cos 100t$ mA.

...erminal voltage across the inductor is:

.....

...The Energy stored in the inductor is:

.....

Prob 2: For questions 3,4,5,6 consider the circuit shown in figure 1. The switch has been closed for a long time. At t=0, the switch is opened. Answer:

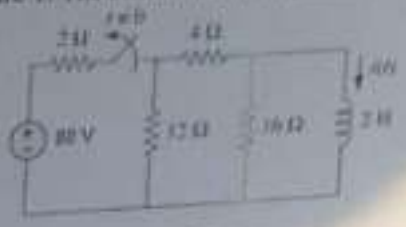


Fig.1

3) The initial current $i(t=0+)$ is:

.....

4) The time constant of the circuit is :

.....

5) The current in the inductor at t= 500 ms is:

.....

6) The final value of the voltage $v(t=\infty)$ across the inductor is :

.....

Prob. 3: For the questions 7,8,9,10 consider the circuit shown in figure 2 and answer:

7) The value of V_o is:

.....

8) The value of the current (i) in 5 Ω is:

.....



Fig.2

9) The power absorbed by 3 Ω resister is:

.....

10) The power absorbed by 12 V voltage source is:

.....

Prob. 4: Consider the circuit shown in figure 3. Use Thevenin's theorem with respect to terminals (a - b) and answer:

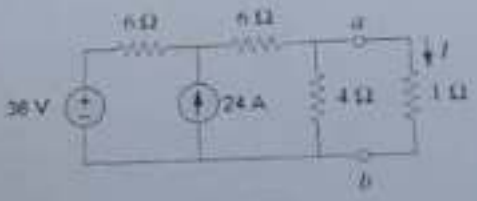


Fig. 3

a. The Thevenin voltage is .....

b. The Thevenin resistance is.....

c. The current I is.....

d. What is the value of the load resistance that should be connected to the ab terminals to achieve maximum power transfer?

.....

a) Refer to figure 4; the value of the equivalent ...ance is:

.....

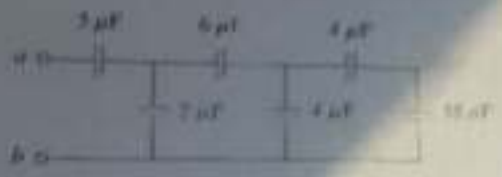


Fig. 4

b) Refer to figure 5; the value of the equivalent resistance is:

.....



Fig. 5

Good Luck

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